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The 4-way deterministic tiling problem is undecidable

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ABSTRACT

It is shown that the (infinite) tiling problem by Wang tiles is undecidable even if the given tile set is deterministic by all four corners, i.e. a tile is uniquely determined by the colors of any two adjacent edges. The reduction is done from the Turing machine halting problem and uses the aperiodic tile set of Kari and Papasoglu.

1. Introduction

A Wang tile (or simply tile) is a unit square tile with colored edges and a Wang tile set is a finite set of Wang tiles. A tiling by Wang tiles is a mapping which assigns a unique tile from the given tile set for each location of the plane. A tiling is said to be valid if the colors of neighboring tiles match. The tiling problem of Wang tiles (also known as the domino problem) is the decision problem of determining whether a given finite tile set admits at least one valid tiling. It was shown by Berger [1] that the tiling problem is undecidable. A simplified proof was given later by Robinson [2]. Both the proof of Berger and the proof of Robinson relied on the existence of an aperiodic Wang tile set, i.e. a Wang tile set which admits only non-periodic valid tilings.

A Wang tile set is said to be 4-way deterministic if for a given pair of colors and a given corner at most one tile in the given tile set is such that the colors of the color pair occur on the tile's edges which are adjacent to the given corner. This definition makes a Wang tile set "deterministic" in all four diagonal directions and the rest of a valid tiling is always determined by a single infinite diagonal row of tiles to any diagonal direction. If one is given an infinite diagonal row of tiles which is already known to be part of a valid tiling, then the rest of the tiling is determined in a unique way.

In this work it is shown that the tiling problem of Wang tiles remains undecidable even if the structure of the given Wang tile set is restricted in the sense that it is 4-way deterministic (Theorem 7) and even if at most one kind of Wang tile can be placed between any two Wang tiles (Theorem 8).

It has been shown earlier that the tiling problem is undecidable when the given Wang tile set is deterministic by one corner [3]. From that fact it follows that the nilpotency of one-dimensional cellular automata is an undecidable problem. It is known that 4-way deterministic aperiodic tile sets exist [4] but it has not been known whether the tiling problem is undecidable or not in the 4-way deterministic case. The connection between Wang tiles and cellular automata and their use in mathematical self-assembly will be briefly reviewed in Sections 2.2 and 2.3, respectively.

2. Preliminaries

2.1. Wang tiles

A *Wang tile* (or a *tile* in short) is a unit square with colored edges. The edges of a Wang tile are called *north*, *east*, *west* and *south* edges in a natural way. Each edge of a Wang tile has a *color* which is an element from a finite set. For the given tile t,

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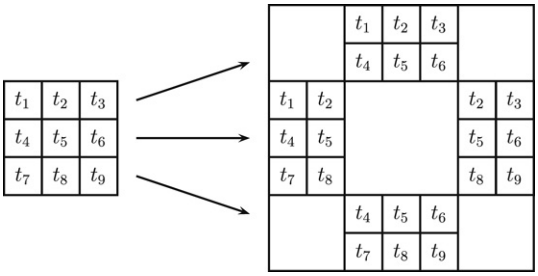

Fig. 1. An $n \times n$ square (or rectangular) cluster of matching Wang tiles can be converted into a single Wang tile. This can be done by setting the $n - 1$ outermost columns and rows of tiles as colors of the new tile.

expressions t_N, t_E, t_W and t_S are used to denote north, east, west and south edge colors, respectively. A *Wang tile set* T (or a *tile set* in short) is a finite set containing Wang tiles.

Given tile sets $T_1, \ldots, T_n$, a tile set $T \subseteq T_1 \times \cdots \times T_n$ is a *sandwich tile set*. Elements of T are called *sandwich tiles*. Tile set T_i is said to be *layer* i of the sandwich tile set $T \subseteq T_1 \times \cdots \times T_n$. Let $t \in T$ be an element of a sandwich tile set and $t = (t_{i_1}, \ldots, t_{i_n})$. Then the colors of t are sequences of corresponding colors of the original tiles, for example, $t_N = (t_{i_1 N}, \ldots, t_{i_n N})$. Let S_i be a subset of tile set T_i for any $1 \le i \le n$. If $S_1 \times \cdots \times S_{i-1} \times \{t\} \times S_{i+1} \times \cdots \times S_n \in T$ it is said that the tile t on layer i is *paired* with tiles S_j on layer j.

Sandwich tiles are perhaps a more illustrative way to express that a tile set is a subset of a cartesian product of given tile sets.

A *tiling* is a mapping $f : \mathbb{Z}^2 \to T$, which assigns a unique Wang tile for each integer pair of the plane. A tiling f is said to be *valid*, if for every pair $(x, y) \in \mathbb{Z}^2$ the tile $f(x, y) \in T$ matches its neighboring tiles (e.g. the south edge of tile $f(x, y)$ has the same color as the north edge of tile $f(x, y - 1)$).

A tiling $f : \mathbb{Z}^2 \to T$ is called *periodic* with period (a, b) if $f(x, y) = f(x + a, y + b)$ for all $(x, y) \in \mathbb{Z}^2$ and $(a, b) \neq (0, 0)$. Otherwise the tiling f is called *non-periodic*. A tile set T is called *aperiodic*, if there exists some tiling with the tile set T, but no tiling with the tile set T is periodic. If the tile set T admits a periodic tiling $f : \mathbb{Z}^2 \to T$ with some period, then it admits also a *doubly periodic* tiling $g : \mathbb{Z}^2 \to T$, that is, there exists such non-zero integers a and b that $g(x, y) = g(x + a, y)$ and $g(x, y) = g(x, y + b)$ for all $(x, y) \in \mathbb{Z}^2$ [2].

A Wang tile set T is said to be *NW-deterministic*, if within the tile set there does not exist two different tiles with the same colors on the north and west edges. In general, a Wang tile set is *XY-deterministic*, if the colors of X- and Y-edges uniquely determine a tile in the given Wang tile set. A Wang tile set is *4-way deterministic*, if it is NE-, NW-, SE- and SW-deterministic.

Let T be a Wang tile set. The $n \times n$ *tile set* is the Wang tile set that has been constructed by taking every $n \times n$ cluster of matching tiles and mapping them to a single Wang tile as shown in Fig. 1. The colors in the vertical direction are all the matching $n \times (n - 1)$ clusters of tiles and the colors in the horizontal direction are all the matching $(n - 1) \times n$ clusters of tiles. The tile set which is the $n \times n$ tile set of the tile set T is denoted by $T^{n \times n}$. Clearly, if the original tile set T is 4-way deterministic so is the tile set $T^{n \times n}$.

A mapping $f : T_1 \to T_2$ is called a *tile homomorphism* if it respects the colors, i.e. $f(t) = t'$ with $t'_N = g(t_N)$, $t'_E = g(t_E)$, $t'_W = g(t_W)$ and $t'_S = g(t_S)$, where g is a mapping from the set of the colors of the tile set T_1 to the set of the colors of the tile set T_2. The *homomorphic image* $f(T)$ of a tile set T is defined in the natural way as the set

$$f(T) = \{f(t) | t \in T\}.$$

If every tile in a given tile set T appears in a valid tiling, then the $n \times n$ tile set $T^{n \times n}$ can always be mapped tile homomorphically onto T, that is, there always exists such a tile homomorphism f that $f(T^{n \times n}) = T$. For example, f could map an element $t \in T^{3 \times 3}$ to the tile t_5 (denoted as in Fig. 1) in the center of the 3×3 tile cluster represented by t.

The following decision problem is referred to as the *tiling problem*: "Given a Wang tile set T, does there exist a valid tiling of the plane?" A tiling $f : \mathbb{Z}^2 \to T$ is said to *contain* tile $t \in T$, if for some integers $x, y \in \mathbb{Z}$ equation $f(x, y) = t$ holds. The following decision problem is referred to as the *tiling problem with a seed tile*: "Given a Wang tile set T and a tile $t \in T$, does there exist a valid tiling of the plane that contains the tile t?"

If the tiling problem with a seed tile were decidable, then the tiling problem would be decidable. Let T be the tile set of the given instance of the tiling problem. Then the answer for the tiling problem is affirmative if, and only if, for some tile $t \in T$ the answer for the tiling problem with a seed tile is affirmative considering the tile set T as the tile set of the instance and the tile t as the seed tile of the instance.

It is already known, that the tiling problem is undecidable [2,1]. Furthermore, it is known to be undecidable even when restricted to tile sets that are deterministic by one corner [3]. In this article it is shown that the tiling problem is undecidable for tile sets that are deterministic by all four corners. The proof relies on the 4-way deterministic aperiodic tile set given by Kari and Papasoglu [4].

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2.2. Determinism in relation to cellular automata

Wang tiles have been used previously, for example, to prove undecidability of injectivity and surjectivity of two-dimensional cellular automata [5] and nilpotency of one-dimensional cellular automata [3]. The definition of determinism for tile sets was originally motivated by the theory of one-dimensional cellular automata [3,6,7].

A tile set can be considered as a one-dimensional cellular automaton, if the tile set contains all the possible color pairs at one corner and the tile set is deterministic by the same corner. The tiles can be seen as states of the cells and the diagonal rows of tiles as configurations of the cellular automaton. Therefore, the rule, which determines whether neighboring tiles match, can in this case be considered as the local rule of a cellular automaton.

It has been shown that the tiling problem is undecidable with tile sets that are deterministic by at least one corner [3]. From this it follows that nilpotency of one-dimensional cellular automata is undecidable [3]. Undecidability of nilpotency follows from the fact that a tiling error can be used to create a spreading quiescent state in the local rule and then every diagonal row of tiles leads to a quiescent configuration if, and only if, there exists no valid tiling.

If a Wang tile set is deterministic by two opposite corners, then the tile set can be seen as a subset of the state set of a reversible one-dimensional cellular automaton. In fact, knowing that the tiling problem is undecidable for *2-way deterministic* tile sets (i.e. tile sets that are deterministic by two opposite corners), one can prove certain properties of reversible one-dimensional cellular automata to be undecidable [7]. From the tiling problem itself it follows that it is undecidable if for any initial configuration some cell always enters a "forbidden" state representing a tiling error. It is also undecidable if for any initial configuration every cell eventually enters a forbidden state. Using a reduction from the tiling problem of 2-way (or 4-way) deterministic tile sets, it can also be shown that a weaker form of expansivity, left expansivity (or right expansivity), is undecidable [7].

2.3. Mathematical self-assembly

Mathematical self-assembly is the concept of modeling chemical self-assembly with Wang tiles. One model for self-assembly is the *tile assembly model* [8], where one is given a tile set T, a *seed tile* $s \in T$, a *temperature* $\tau \in \mathbb{N}$ and a *glue function* $g : C \times C \to \mathbb{Z}$, where C is the set of colors of T and $g(a, b) = g(b, a)$. The tiles represent molecules and the glue function is used to represent the bond strength between the edges of different molecules (i.e. tiles). In its simplest form, self-assembly is started from a single seed tile and tiles are added one by one to an existing connected cluster of tiles. A tile can be appended to an existing tile cluster if there is a slot around which the sum of the glue values between the tile and the earlier tiles exceeds or equals the temperature. A more thorough discussion of mathematical self-assembly can be found, for example, in [8–10].

One topic of mathematical self-assembly is the characterization of tile sets (and glue functions) that produce desired kind of tile clusters even if any finite number of tiles (except the seed tile) were removed from the tile cluster. For this reason, self-healing tile sets have been introduced in [11,12]. The approach is to counter a sudden removal of any finite number of tiles (except the seed tile) from assembled tile clusters. A tile set is called *self-healing* if for any assembled tile cluster any damage caused by removing any finite number of tiles is eventually repaired (in linear time with respect to the number of removed tiles) so that every removed tile is restored in its original place in the tile cluster.

One possible application of 4-way deterministic tile sets is the modeling of one type of self-healing systems in mathematical self-assembly. In a way, 4-way deterministic tile sets are an analogy of self-healing tile sets in the case of classical tilings. If a 4-way deterministic tile set admits a valid tiling, then any continuous two-way infinite tile path, which intersects all rows and columns and is contained in a valid tiling, determines rest of the tiling uniquely. If any finite number of tiles is removed from a valid tiling, the missing tiles can be replaced only in a unique way to produce a valid tiling again. Any two adjacent neighbors determine a tile uniquely in a valid tiling. With regard to mathematical self-assembly, Theorem 8 gives an even more useful result. It states that the tiling problem is undecidable even when the tile set is deterministic by any two, including opposite, edges. This follows from the undecidability of the tiling problem in the 4-way deterministic case and the fact that a 4-way deterministic tile set can be changed to a 2×2 tile set which is deterministic also with respect to colors on opposite edges of tiles.

2.4. Turing machines

In what follows, the reader is assumed to be somewhat familiar with the concept of a Turing machine [13,14]. In this article, a *Turing machine* $\mathcal{M}$ is considered to be a four-tuple $\mathcal{M} = (Q, \Gamma, \delta, q_0)$, where Q is the state set, Γ is the tape alphabet, δ is the transition function and $q_0 \in Q$ is the initial state. No "accept", "reject" or "halt" states are defined explicitly. The tape of a Turing machine is defined to be two-way infinite and symbol ε is used to denote the empty symbol of the Turing machine. The transition function is a partial mapping

$$\delta : Q \times \Gamma \to Q \times \Gamma \times \{\triangleleft, \triangleright\},$$

that is, at every time step the read-write head moves either to the left or to the right or halts. A Turing machine is said to *halt*, if it is in state q reading symbol s and $\delta(q, s)$ is undefined. The *Turing machine halting problem* is the following decision problem: "Does the given Turing machine $\mathcal{M}$ halt when started on an empty tape?" The halting problem is known to be undecidable.

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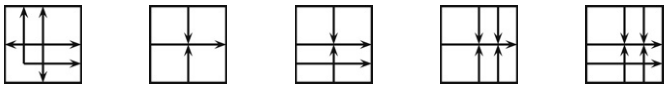

(a) A cross. (b) An arm. (c) An arm. (d) An arm. (e) An arm.

Fig. 2. The basic tiles of Robinson's tile set (with colors, reflections, rotations and parity constraints omitted).

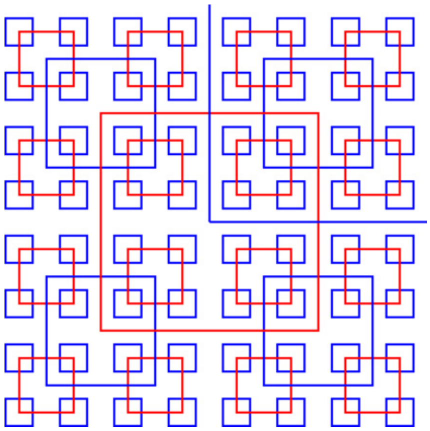

Fig. 3. A part of the self-similar pattern generated by the tile set of Robinson (and the tile set of Kari and Papasoglu). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3. The aperiodic tile set

3.1. The aperiodic tile set of Robinson

Earlier proofs of the undecidability of different variants of the tiling problem have relied heavily on the existence of aperiodic tile sets, that is, tile sets which admit only non-periodic valid tilings. A well-known aperiodic tile set is the tile set of Robinson which he used in proving the tiling problem to be undecidable [2]. Its 4-way deterministic version will be used in the following proofs also.

A general outline of Robinson's tile set is shown in Fig. 2. The tile set consists of tiles that are called *crosses* and tiles that are called *arms* as shown in the figure. The colors of the tiles are defined using patterns consisting of *single arrows* and *double arrows*. The arrows are colored either red or blue. In a cross tile all the arrows have the same color. In an arm tile the horizontal arrows may have the same or a different color than the vertical arrows with the exception that the tiles of the form shown in Fig. 2(e), where the horizontal arrows are required to have a different color than the vertical arrows. This constraint concerning the tiles of the form shown in Fig. 2(e) is set to ensure that only squares with different color can intersect.

Two tiles are considered to match at their abutting edges if an arrow (of some particular type and color) exiting one of the tiles enters the other tile. Robinson's tile set has also some parity constraints that are not shown in the tiles in Fig. 2. A more thorough description of Robinson's tile set can be found in [2].

Robinson's tile set forces a self-similar pattern to be tiled, a part of which is shown in Fig. 3. The tiling forced by the Robinson's tile set is divided into square areas bounded by blue squares or red squares. The edges of the squares are formed from the double arrows found in the tiles in Fig. 2. The crosses (i.e. the tiles of the form in Fig. 2(a)) act as corners for the squares of different size and color.

More specifically, the tiling contains blue squares of height $2^{2n+1} + 1$ and red squares of height $2^{2(n+1)} + 1$, for every non-negative integer n. Furthermore, the borders of the squares of the same color never coincide. In the center of the area bounded by a blue square there is always some corner of a red square, and likewise in the center of the area bounded by a red square there is always some corner of a blue square. However, every blue cross is not located in the center of a red square because it is assumed that the smallest squares present in a valid tiling are the blue squares.

3.2. The 4-way deterministic aperiodic tile set

Kari and Papasoglu have constructed a 4-way deterministic tile set which will be used in this article instead of Robinson's tile set. Formally, the following theorem holds:

Theorem 1 (*Kari and Papasoglu [4]*). *There exists a 4-way deterministic tile set, which*

(1) *admits a valid tiling and*
(2) *can be mapped homomorphically onto Robinson's tile set.*

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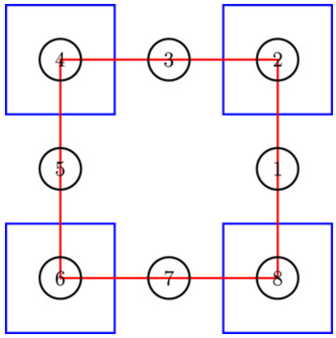

Fig. 4. The tiles around the edges of blue and red squares in the non-periodic tiling which are used to set constraints between the aperiodic tile set and the final sandwich tile set. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

An implication of property (2) is the aperiodicity of the tile set of Kari and Papasoglu. However, the exact structure of the 4-way deterministic tile set is irrelevant. It is entirely sufficient to know that there exists a 4-way deterministic tile set which can be mapped homomorphically onto Robinson's tile set. That is, the tile set can be used in a similar way as Robinson's tile set and it is enough to refer to the 4-way deterministic aperiodic tile set as if it were the Robinson's tile set. The only difference is that each tile in the Robinson's tile set is represented by possibly more than one tile in the tile set of Kari and Papasoglu.

4. Drawing a diagonal line

In this section it is shown how to construct a 4-way deterministic tile set which can be used to draw a single diagonal line. That is, one can construct such a tile set D, that it is a union of disjoint tile sets D_1 and D_2 and it admits a valid tiling in which the tiles of D_1 are located only on a single two-way infinite diagonal row of tiles and no tile of D_2 is located on the same diagonal row.

In practical terms the tile set D is such that if the tiles of D_1 were colored black and the tiles of D_2 were colored white, then there would exist such a valid tiling that it contains a single black diagonal line on white background. However, the tiling which contains the single diagonal line is not forced. There do exist other valid tilings and not just the one having a single diagonal line. Fortunately, it is not necessary to force the diagonal line to be tiled.

The construction is quite tedious but in Section 5.5 it is essential in converting a "nondeterministic" set of tiles which represents a Turing machine into a tile set which is 4-way deterministic.

4.1. Structure of the tile set

The tile set D is constructed in four layers as follows:

Layer 1. The aperiodic tile set of Kari and Papasoglu.
Layer 2. The tiles in Figs. 5–7.
Layer 3. The tiles of layer 1 rotated by 180 deg.
Layer 4. The tiles of layer 2 rotated by 180 deg.

The tile set D is constructed as a sandwich tile set with four layers. However, tiles of layers 1 and 2 will be paired together in a similar way as the tiles of layers 3 and 4 . Therefore it is, for now, sufficient to describe how the tiles of layers 1 and 2 are paired together.

The first layer consists of the tiles of the aperiodic tile set of Kari and Papasoglu and the second layer consists of tiles from a simpler tile set. Each tile of the aperiodic tile set is paired with a specific tile set (one of the sets shown in Figs. 5–7). Recall that in a valid tiling by the Robinson's tile set the squares are formed from the arrow patterns of the tiles in Fig. 2. Then the corners and the edge centers (enumerated as in Fig. 4) of a square can be described as follows in terms of the arrow patterns (which are used as colors in the aperiodic tile set):

(1) A meeting point of two vertical double arrows at the east border of a square.
(2) A cross in the northeast corner of a square.
(3) A meeting point of two horizontal double arrows at the north border of a square.
(4) A cross in the northwest corner of a square.
(5) A meeting point of two vertical double arrows at the west border of a square.
(6) A cross in the southwest corner of a square.
(7) A meeting point of two horizontal double arrows at the south border of a square.
(8) A cross in the southeast corner of a square.

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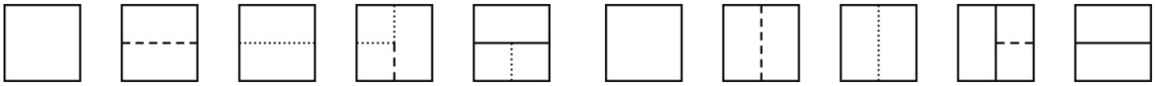
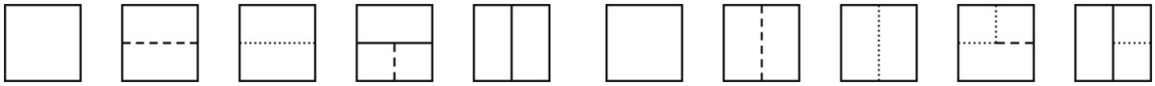
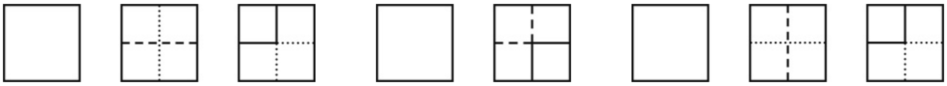


(a) The second layer tiles for tiles of the aperiodic tile set in locations 1. (b) The second layer tiles for tiles of the aperiodic tile set in locations 3.

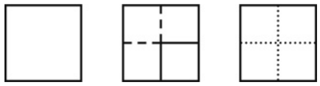


(c) The second layer tiles for tiles of the aperiodic tile set in locations 5. (d) The second layer tiles for tiles of the aperiodic tile set in locations 7.

Fig. 5. The tiles to be paired with the center tiles of blue and red square edges.

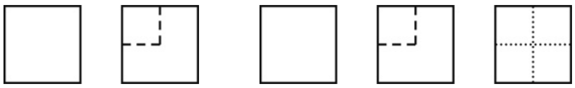


(a) The second layer tiles for tiles of the aperiodic tile set in locations 2. (b) The second layer tiles for tiles of the aperiodic tile set in locations 4. (c) The second layer tiles for tiles of the aperiodic tile set in locations 6.

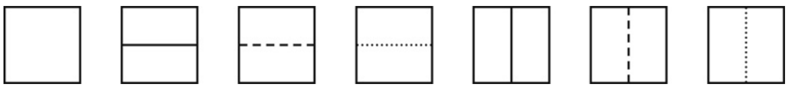


(d) The second layer tiles for tiles of the aperiodic tile set in locations 8.

Fig. 6. The tiles to be paired with the tiles in the corners of blue and red squares.



(a) The second layer tiles for tiles of the aperiodic tile set in locations 4 instead of the tiles in Fig. 6(b) if the tile of the first layer is in the center of a 3×3 blue square. (b) The second layer tiles for tiles of the aperiodic tile set in locations 8 instead of the tiles in Fig. 6(d) if the tile of the first layer is in the center of a 3×3 blue square.



(c) The tiles of layer 2 to be paired with such tiles of layer 1 that are not located in any of the locations in Fig. 4.

Fig. 7. The tiles to be paired with the tiles in the center of the smallest blue squares and in other locations than the ones in Fig. 4.

It is possible to assume that one can make a distinction between the tiles that are located within a square of a certain fixed size and the tiles that are not. This can be done by forming a new tile set by taking sufficiently large $n \times n$ blocks (which do not contain tiling errors) of the original tiles and using $n - 1$ outermost tile rows and columns of tiles of the blocks as colors of the new tiles as in Fig. 1. This construction maintains 4-way determinism. Therefore it is possible to assume that one can determine from the edge colors whether a given tile from the aperiodic tile set is located inside a blue 3×3 square in a valid tiling. Now the tiles on layer 1 and layer 2 are paired together according to the following rules:

(1) Tiles at locations 1 on layer 1 are paired with the tiles in Fig. 5(a) on layer 2 .
(2) The tiles at locations 2 on layer 1 are paired with the tiles in Fig. 6(a) on layer 2 .
(3) The tiles at locations 3 on layer 1 are paired with the tiles in Fig. 5(b) on layer 2 .
(4) The tiles at locations 4 on layer 1 which are located outside a 3×3 blue square are paired with the tiles in Fig. 6(b) on layer 2 . The tiles at locations 4 on layer 1 which are located within a 3×3 blue square are paired with the tiles in Fig. 7(a) on layer 2.
(5) The tiles at locations 5 on layer 1 are paired with the tiles in Fig. 5(c) on layer 2.
(6) The tiles at locations 6 on layer 1 are paired with the tiles in Fig. 6(c) on layer 2 .
(7) The tiles at locations 7 on layer 1 are paired with the tiles in Fig. 5(d) on layer 2 .

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(8) The tiles at locations 8 on layer 1 which are located outside a 3×3 blue square are paired with the tiles in Fig. 6(d) on layer 2 . The tiles at locations 8 on layer 1 which are located within a 3×3 blue square are paired with the tiles in Fig. 7(b) on layer 2.
(9) Other tiles than the ones in locations shown in Fig. 4 are paired with the tiles in Fig. 7(c).

This new tile set is almost 4-way deterministic. All the tile sets in Figs. 5(a)–(d), 6(a)–(d) and 7(c) are 4-way deterministic. However, the tile sets in Fig. 7(a) and (b) cause the sandwich tile set to be not 4-way deterministic. Fortunately, these are the only sources of "nondeterminism".

Let T denote the newly constructed tile set with two layers. Let A denote the 3×3 tile set of the aperiodic tile set of Kari and Papasoglu. Let B denote the union of all tiles in Figs. 5–7. Then

$$T \subseteq A \times B.$$

Let $A_1 \subseteq A$ denote the tiles in locations 2 or 6 in the center of a 3×3 blue square. Let t_d denote the second tile found in the tile sets in Fig. 7(a) and (b). Let $T_1 = A_1 \times \{t_d\}$ and $T_2 = T \setminus T_1$. Notice that no sandwich tile of T_2 contains the tile t_d because by definition t_d is paired only with the tiles of A_1.

Now the tile set U_1 (which is used to draw a "dotted" diagonal line) is defined to be

$$U_1 = T_1 \times T_1^{\circlearrowleft}, \tag{1}$$

where $T_1^{\circlearrowleft}$ denotes the tile set which contains the tiles of T_1 after they have been rotated by 180 deg. The tile set U_2 (which represents the background color) is defined to be

$$U_2 = T_2 \times T_2^{\circlearrowleft},$$

where $T^{\circlearrowleft}$ denotes the tile set which contains the tiles of T after they have been rotated by 180 deg. Now the tile set U is defined to be

$$U = U_1 \cup U_2.$$

This definition of the tile set U sets a constraint in Eq. (1) that on layer 2 the third tile in Fig. 7(a) and (b) is used if, and only if, its rotated counterpart is used on layer 4. This will make the tile set 4-way deterministic because the tile t_d (the second tile Fig. 7(a) and (b)), which causes the nondeterminism, is paired only with its rotated counterpart (which will be denoted by $t_d^{\circlearrowleft}$). The tile t_d cannot be distinguished from the blank tile by looking at the colors on the southeast corner whereas the tile $t_d^{\circlearrowleft}$ can be distinguished from the blank tile by looking at the colors on the southeast corner. To say it concisely, the union U of tile sets

$$U_1 = A_1 \times \{t_d\} \times A_1^{\circlearrowleft} \times \{t_d^{\circlearrowleft}\} \quad \text{and}$$
$$U_2 = T_2 \times T_2^{\circlearrowleft}$$

is 4-way deterministic because the elements of U_1 can be distinguished from the elements of U_2 by watching the colors adjacent to any corner because no sandwich tile of T_2 contains the tile t_d.

Lemma 2. *The tile set U is 4-way deterministic.*

4.2. Drawing the diagonal recursively

In this section it is described how the line patterns drawn by the non-blank tiles in Figs. 5–7 can be used to distinguish the tiles in a single infinite diagonal row from the tiles in the rest of the tiling. The idea is that the tile set D is such that it admits a tiling which contains a fractal-like line pattern, part of which is shown in Fig. 8. For brevity, the construction is described again only on layers 1 and 2 because for layers 3 and 4 it is similar.

The most important observation concerning the line pattern in Fig. 8 is that the upper left half of it can be partitioned into smaller line patterns shown in Fig. 9(a) and (b). The other half can also be partitioned into smaller line patterns which are the ones in Fig. 9(a) and (b) after rotating by 180 deg. It is enough to show that the upper left half of the pattern shown in Fig. 8 can be drawn with the tiles of layer 2 of tile set T because the same pattern can be repeated on layer 4 as a reflection.

With respect to Fig. 8, Fig. 9(a) and (b) should be interpreted so that a line pattern starting from a location denoted by a filled circle is repeated with (approximately) half the size in locations denoted by the empty circles. The line pattern is repeated in smaller size over and over again until the locations denoted by the empty circles are found in the centers of the 3×3 blue squares in which case the pattern is no longer repeated.

Lemma 3. *The tile set T admits such a valid tiling $f : \mathbb{Z}^2 \to T$ that $f(x, y) \in T_1$ if, and only if, $x = y$ and $x, y \in 4\mathbb{Z}$.*

Proof. First, it can be noted (by listing all the possibilities) that all the different "line interactions" seen in Fig. 9(a) and (b) in the locations in Fig. 4 have been implemented as the tiles in Figs. 5 and 6. Furthermore, the tiles used elsewhere than in locations 4 are exactly the tiles in Fig. 7.

Second, it can be concluded with a simple inductive argument that the recursive pattern in Fig. 9(a) and (b) can be drawn for arbitrary depth without tiling errors and therefore the tiling f exists (Fig. 10). □

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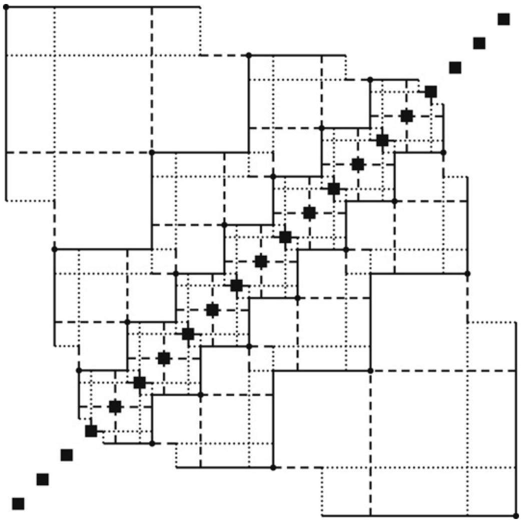
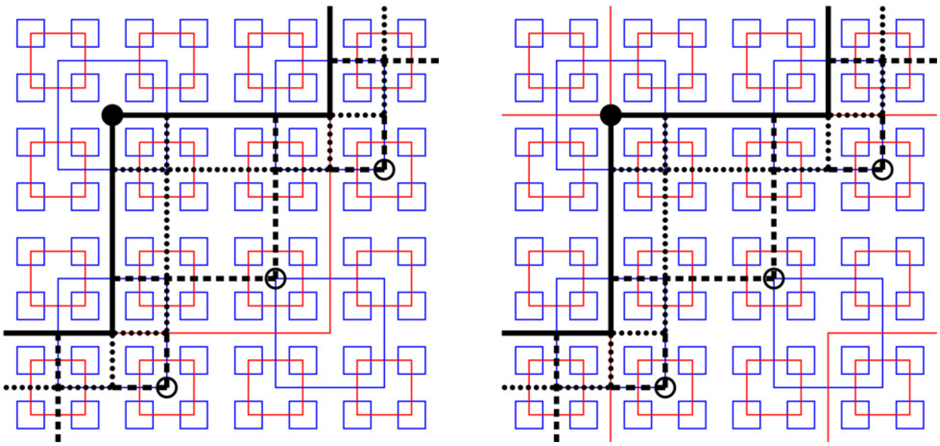


Fig. 8. The structure of the tile set D is such that the line patterns on top of the non-periodic tiling can draw a recursive line pattern so that a single diagonal row of tiles is tiled with a different set of tiles than rest of the tiling. The locations of sandwich tiles containing tile t_d (and its rotated counterpart) are denoted by black squares.



(a) A line pattern starting from the top left corner of a square.

(b) A line pattern starting from the bottom right corner of a square.

Fig. 9. A recursive signal pattern is used to draw a diagonal line. The partial pattern beginning at the location denoted by a filled circle is recursively repeated with half the previous size at the locations denoted by empty circles.

Lemma 4. *The* 4*-way deterministic tile set U admits such a valid tiling $f : \mathbb{Z}^2 \to U$ that $f(x, y) \in U_1$ if, and only if, $x = y$ and $x, y \in 4\mathbb{Z}$.*

Proof. Follows directly from the previous lemma because the tiles of T_1 and $T_1^{\circlearrowleft}$ are paired together which makes the tile set 4-way deterministic. □

Let D denote the 4×4 tile set of the tile set U, that is, $D = U^{4\times 4}$. Consider the 4×4 tiles as 4×4 tile clusters in the sense of Fig. 1. Then D_1 is the subset of D containing the tiles that have an element of U_1 located somewhere on the central diagonal (with positive slope). Tile set D_2 is the subset of D containing the tiles that do not have an element of U_1 located anywhere on the central diagonal. Clearly, $D = D_1 \cup D_2$.

Theorem 5. *There exists such a tile set $D = D_1 \cup D_2$ (where $D_1 \cap D_2 = \emptyset$) that it is* 4*-way deterministic and there exists such a valid tiling $f : \mathbb{Z}^2 \to D$ that $f(x, y) \in D_1$ if, and only if, $x = y$.*

Proof. Follows directly from the previous lemma because switching to a 4×4 tile set does not remove the 4-way determinism but enables to determine whether a given tile from U_2 is located diagonally between two tiles from U_1 in the tiling f given by the previous lemma. □

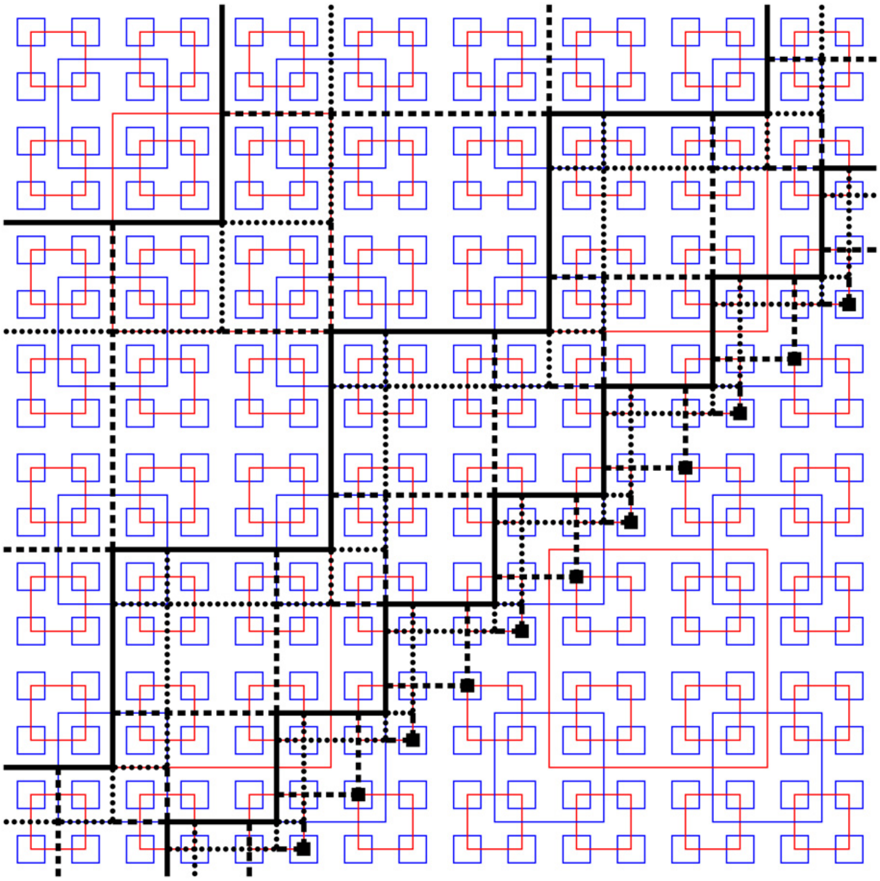


Fig. 10. The location of the diagonal line is determined by tiles (denoted by small black squares in the picture) which are located in the centers of the 3×3 blue squares and receive dashed lines through their north and west edges. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5. The tiling problem with a seed tile

In this section it will be shown that the tiling problem with a seed tile remains undecidable even if the instances are 4-way deterministic tile sets. This can be seen by applying an additional construction represented in Section 4 to the original tile set of Robinson.

5.1. The idea for the undecidability proof

The basic idea is to represent Turing machine configurations on horizontal tile rows as was already done in [15]. One tile at each row represents the read-write head and the current symbol to be read. The rest of the tiles on the same row represent other cells of the tape located to the left and to the right from the read-write head. This tile set construction has been used earlier in [15,2].

Unfortunately, the tile set construction does not provide a 4-way deterministic tile set. However, the tile set can be modified to make it 4-way deterministic by using the construction given in Section 4. After the modification, the undecidability with the 4-way deterministic instances follows directly.

Formally, the tile set is constructed (as a sandwich tile set) in four layers as follows:

Layer 1. The tiling representing a Turing machine computation.
Layer 2. Horizontal signals identifying the move tile pair (on layer 1) occurring on the particular row.
Layer 3. The tiles used to distinguish the leftmost move tiles from symbol tiles of layer 1.
Layer 4. The tiles used to distinguish the rightmost move tiles from symbol tiles of layer 1.

Eventually, the following theorem holds:

Theorem 6. *The tiling problem with a seed tile remains undecidable when restricted to tile sets that are 4-way deterministic.*

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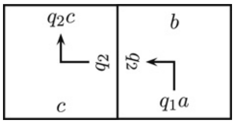
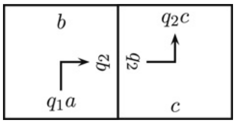
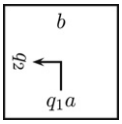


(a) The left move $\delta(q_1, a) = (q_2, b, \triangleleft)$.

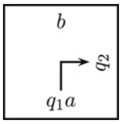


(b) The right move $\delta(q_1, a) = (q_2, b, \triangleright)$.

Fig. 11. The tile combinations to represent different Turing machine moves.

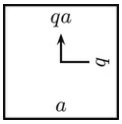


(a) The action tile for the left move $\delta(q_1, a) = (q_2, b, \triangleleft)$.

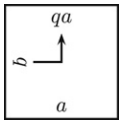


(b) The action tile for the right move $\delta(q_1, a) = (q_2, b, \triangleright)$.

Fig. 12. Action tiles.



(a) Merging tiles for all the left moves.



(b) Merging tiles for all the right moves.

Fig. 13. Merging tiles.

Proof. Section 5.2: The tiles of layer 1 can be used to represent an unbounded Turing machine computation started on a blank tape. This layer is forced to contain a tiling error if, and only if, the Turing machine halts.

Section 5.4: The tile set can be modified to have no ambiguity between different action tiles or merging tiles at any corner. The tiles of layer 2 are used to distinguish one horizontal pair of an action tile and a merging tile from other tile pairs. This is done by choosing a unique color for any pair of move tiles. The new color is transferred horizontally and paired only with the particular move tiles. Hence, no two move tiles belonging to different read-write operations have the same color on their east edges or their west edges. This layer can be tiled in a valid way if the tiling on layer 1 is valid.

Section 5.5: The tile set can be modified to have no ambiguity between an action tile, a merging tile or a symbol tile at any corner. The tiles of layer 3 are used to distinguish the leftmost tile of the tile pair representing a move from the symbol tiles of the same row. Likewise, the tiles of layer 4 are used to distinguish the rightmost tile of the tile pair representing a move from the symbol tiles of the same row. These layers can be tiled in a valid way if the tiling on layer 1 is valid.

The tile sets used on layers 2, 3 and 4 are all 4-way deterministic, so they do not create nondeterminism. The way they are paired with the original tile set on layer 1 makes the final tile set 4-way deterministic. □

Details of the proof are given in the Sections 5.2–5.5.

5.2. Layer 1: Simulating an unbounded Turing machine computation

In this subsection a tile set used for representing a Turing machine computation will be presented [15]. The tiles are used on layer 1 to represent a Turing machine computation which is started from a blank tape.

Left moves Assume that the Turing machine contains move $\delta(q_1, a) = (q_2, b, \triangleleft)$. Then the tile combination in Fig. 11(a) (for every symbol c) is used to represent the move. Therefore the tile in Fig. 12(a) and all the tiles in Fig. 13(a) are added to the tile set.

Right moves Assume that the Turing machine contains move $\delta(q_1, a) = (q_2, b, \triangleright)$. Then the tile combination in Fig. 11(b) (for every symbol c) is used to represent the move and the tile in Fig. 12(b) and all the tiles in Fig. 13(b) are added to the tile set.

Tape contents For every tape alphabet element a, a tile of the form in Fig. 14 is added to the tile set. This tile represents a single tape cell and its current contents.

Initial configuration To force the Turing machine to start on a blank tape only, the tiles in Fig. 15 are added to the tile set. One of these tiles (namely, the one in Fig. 15(b)) is chosen to be the seed tile. If the seed tile is contained within a

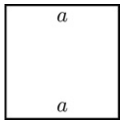
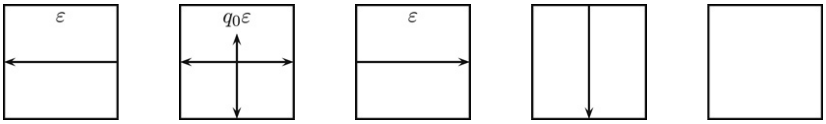


Fig. 14. An alphabet tile.



(a) Left side initial tape. (b) The seed tile. (c) Right side initial tape. (d) Downward extension for the seed tile. (e) The blank tile.

Fig. 15. Starting tiles.

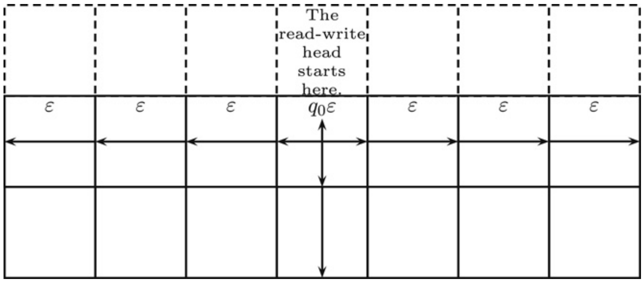


Fig. 16. The tile pattern representing the initial configuration of the Turing machine computation.

tiling, then a valid tiling will necessarily represent a non-halting Turing machine computation. The tiles in Fig. 15(a) and (c) define the tape to be initially empty. In short, if the seed tile is located in the origin, then the Turing machine simulation is done in the first and the second quadrant of the plane. The tiles in Fig. 15(d) and (e) allow the lower part of the plane to be always correctly tiled. An example is given in Fig. 16 on the use of tiles in Fig. 15.

Following the terminology of Robinson [2], the tiles shown in Fig. 12 are referred to as *action tiles*. Every Turing machine move is represented by a unique action tile. The tiles shown in Fig. 13 are called *merging tiles* and the tiles of the form in Fig. 14 are called *alphabet tiles*. The tiles in Fig. 15 are referred to as *starting tiles*. The final tile set consists of a unique action tile for every non-halting move, all the possible merging tiles, all the possible alphabet tiles and the starting tiles.

Let q and a be a pair of a read-write head state and a tape symbol for which the Turing machine transition $\delta(q, a)$ has not been defined. Then there will be no tile that would have the color qa on its south edge. Therefore, if the Turing machine halts, that is, if at some moment of time the read-write head in state q reads symbol a, then the tiling cannot be completed to cover the entire plane in a valid way. From this it follows that the simple version of the tiling problem with a seed tile is undecidable.

5.3. Nondeterminism of the Turing machine tiles

The Turing machine tiles are themselves sufficient to show the original tiling problem with a seed tile to be undecidable. However, the tile set is not 4-way deterministic.

The tile set is not 4-way deterministic (only) because

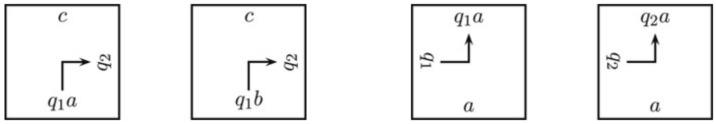
Problem 1. two different action or merging tiles cannot always be distinguished from each other (see Fig. 17 for the case of right moves) and

Problem 2. action and merging tiles cannot always be distinguished from the symbol tiles (see Fig. 18 for the case of right moves).

Fortunately, these are the only sources of nondeterminism in the tile set. The tiling problem (with a seed tile) will be seen to be undecidable in the 4-way deterministic case by modifying the Turing machine tiles so that problems 1 and 2 are solved (and the resulting tile set is 4-way deterministic otherwise also). Problem 1 will be removed in Section 5.4 and problem 2 will be removed in Section 5.5.

5.4. Layer 2: Distinguishing different move tiles

Let the number of different action tiles and merging tiles of layer 1 be n for the given Turing machine. For simplicity, let the different action tiles and merging tiles be denoted by expressions $t_1, \ldots, t_n$, that is, integer i identifies tile t_i uniquely.



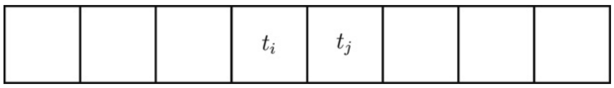
(a) Ambiguity at the northwest corner. (b) Ambiguity at the southeast corner.

Fig. 17. The tiles representing a Turing machine move cannot be distinguished.

(a) Ambiguity at the northwest corner. (b) Ambiguity at the southeast corner.

Fig. 18. A tile representing a Turing machine move and an alphabet tile cannot be distinguished.

Fig. 19. The tiles of layer 2.



(a) Layer 1.

(b) Layer 2.

Fig. 20. Associating tiles t_i and t_j with the horizontal signal identifying the tile pair.

Let t_k be any action tile or merging tile occurring on layer 1. Then the tile is paired with the tiles of the form in Fig. 19 with either $i = k$ or $j = k$. If the tile is of the form in Fig. 12(b) or 13(a), it is required that $i = k$ and $j \neq k$. Otherwise, if the tile is of the form in Fig. 12(a) or 13(b), it is required that $i \neq k$ and $j = k$. In other words, in the tile pair representing a move, the leftmost tile is identified by the first component and the rightmost tile is identified by the second component in the pair (i, j). This makes it possible to distinguish any two action tiles or merging tiles from each other just by observing the color of (either) the east edge or the west edge.

All the other tiles are paired freely with all the tiles of the form in Fig. 19, where $1 \leq i, j \leq n$. On a valid tiling, layer 2 consists of rows of tiles like the one in Fig. 20(b) if the underlying tile row contains action tile and move tile pair t_i and t_j as in Fig. 20(a).

5.5. Layers 3 and 4: Distinguishing move tiles from symbol tiles

Let D be the tile set of Theorem 5 which is used to draw a northeast-southwest diagonal line 4-way deterministically. Let $D = D_1 \cup D_2$, where D_1 is the set of tiles used on the diagonal line only and let $D_2 = D \setminus D_1$. Let D^R be the tile set which has been constructed by interchanging the north edge colors and south edge colors in the tiles of set D and by replacing the east edge colors and the west edge color bijectively with a completely new set of colors. Tile set D^R can be used to tile a diagonal line in the northwest-southeast direction. Let D_1^R be the set of tiles located on this diagonal pattern and let again $D_2^R = D^R \setminus D_1^R$.

Now tile set $D \cup D^R$ is 4-way deterministic and it can be used to draw any diagonally advancing zig-zag line with exactly the elements $D_1 \cup D_1^R$ on the zig-zag line. Using tile sets D and D^R one can distinguish action tiles and merging tiles from alphabet tiles by pairing the action tiles and the merging tiles of layer 1 with the tiles representing a diagonal line.

The action tiles in Fig. 12(a) are paired with the tiles of set D_1^R on layer 4. The action tiles in Fig. 12(b) are paired with the tiles of set D_1 on layer 3. The merging tiles in Fig. 12(a) are paired with the tiles of set D_1^R on layer 3. The merging tiles in Fig. 12(b) are paired with the tiles of set D_1 on layer 4. These constraints force two diagonally advancing ziz-zag lines to be drawn side by side on layers 3 and 4. The line pattern of layer 3 is associated with merging tiles of left moves and action

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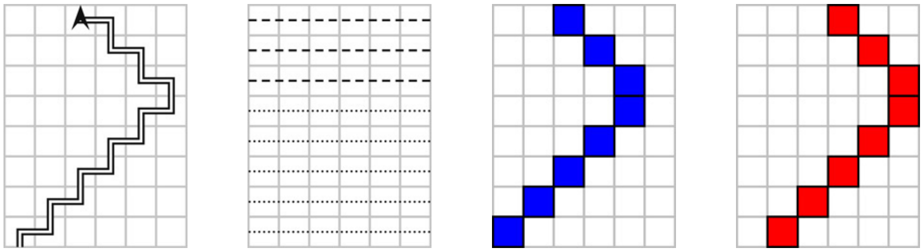


(a) Read-write head movement on layer 1.

(b) Signals on layer 2 for different moves.

(c) Simulating a diagonal line on layer 3.

(d) Simulating a diagonal line on layer 4.

Fig. 21. Distinguishing move tiles from symbol tiles by using diagonal patterns that can be drawn 4-way deterministically.

tiles of right moves (that is, the leftmost tile in a tile pair representing a move). The line pattern of layer 4 is associated with merging tiles of right moves and action tiles of left moves (that is, the rightmost tile in a tile pair representing a move).

All the alphabet tiles are paired with all the tiles in $D_2 \cup D_2^R$ on both layers and the starting tiles are paired freely with all the tiles $D \cup D^R$ on both layers. This can be done because there is no ambiguity between starting tiles and other tiles.

The idea of this construction is better seen in Fig. 21. The left side tiles are identified by the tiles on the diagonal lines on layer 3 and the right side tiles are identified by the tiles on the diagonal lines on layer 4.

If layer 1 has been tiled in a valid way, layers 3 and 4 can also be tiled correctly. This follows from the fact that a row of tiles of D can always be followed by a row of tiles D^R with matching colors since D^R is only a "reflected" version of D and D admits a valid tiling.

This construction distinguishes the action tiles and the merging tiles from the rest of the tiles. By looking at the tiles on layers 3 and 4 it can be determined whether the tile in question is an alphabet tile or not. The tile is an alphabet tile if, and only if, it is not a starting tile and it is not paired with a member of sets D_1 or D_1^R on layers 3 and 4.

6. The tiling problem without a seed tile

In this section the tiling problem without a seed tile is shown to be undecidable even for 4-way deterministic tile sets. The argumentation is quite similar to that of earlier proofs [3,2]. The only difference is the requirement that the final tile set must be 4-way deterministic.

6.1. A brief outline of the argumentation

The reduction is done from the tiling problem with a seed tile to the tiling problem (when restricted to the instances that are deterministic by all four corners, of course). That is, if there were an algorithm for solving the tiling problem, then there would also be an algorithm for solving the tiling problem with a seed tile.

The idea of the reduction is to construct a more complex tile set according to the original tile set. For the new tile set, the answer for the tiling problem will be affirmative if, and only if, the answer for the tiling problem with a seed tile is affirmative for the original given tile set and the given seed tile.

The new tile set is such that certain areas of a valid tiling are used to simulate a tiling with the original tile set. These areas are referred to as free rows and free columns. Identifying the free areas in the earlier case [3] required some modifications to the original proof of Robinson [2]. Now the tile set construction for identifying the free areas is more complicated.

The construction of the new tile set relies heavily on the use of an aperiodic tile set. By using the square patterns generated by Robinson's tile set, copies of the seed tile are forced to be located at certain points of the plane.

In [3], Kari modified Robinson's tile set resulting a new aperiodic tile set which is deterministic by one corner. Later Kari and Papasoglu presented an aperiodic 4-way deterministic tile set which can be mapped homomorphically onto Robinson's original aperiodic tile set [4]. This 4-way deterministic tile set will be used in the proof instead of Robinson's tile set.

The new tile set is constructed in six layers for the given tile set T and a seed tile $t \in T$. The rough outline of the layers is the following:

Layer 1. The tiling forced by the 3×3 tile set (see the definition in Section 2.1) of the aperiodic tile set of Kari and Papasoglu.
Layer 2. The tiles to identify free areas.
Layer 3. A tiling simulating a tiling by the given tile set T.
Layer 4. The tiles to forward the colors of the tile set T on layer 3 from a free area to another free area border and from a red border to another red border without discarding any color information.

Theorem 7. *The tiling problem is undecidable even when restricted to tile sets that are* 4*-way deterministic.*

Proof. Section 6.2: It is possible to divide the plane into squares of increasing size using (the 3×3 tile set of) the aperiodic tile set of Kari and Papasoglu. The squares are colored either red or blue. No edges of two squares of the same color can coincide.

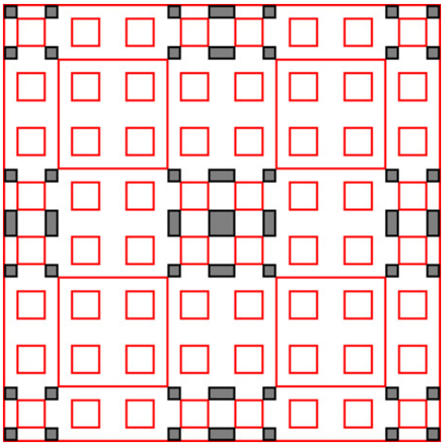


Fig. 22. The free area of 9 × 9 squares within a red square spanning 65 × 65 squares.

Section 6.3: Each of the red squares contains free areas that are not between any of the smaller red squares. The tiles which are used on free and non-free areas belong to two disjoint subsets of the modified aperiodic tile set.

Section 6.4: A finite area of a tiling by the original tile set is simulated on the free areas within the red squares. The size of the free area inside a red square square is directly proportional to the size of the red square. One copy of the seed tile can be forced to be located in the center of every red square (and therefore in the center of every simulation area) by pairing only the seed tile with all blue cross tiles that are not part of a 3 × 3 blue square. Whether a cross tile is part of a 3 × 3 blue square is determined by using a 3 × 3 tile set (see Section 2.1 for the definition) version of the aperiodic tile set.

Section 6.5: The area consisting of disjoint free areas can be considered as a single continuous square. This is seen by transferring the colors between the free areas using a 4-way deterministic construction. Also, if the original tile set admits arbitrarily large squares to be tiled, so does the new tile set. This is possible because the colors next to a red edge on a free area are chosen nondeterministically and shared between all the red squares of the same size using horizontal and vertical signals containing the color information. Then the plane is tiled correctly if, and only if, on every red square the free areas are tiled correctly using the original tile set. Any valid tiling by the original tile set can be simulated using the new tile set without a tiling error. □

The details of the proof of Theorem 7 have been scattered to Sections 6.2–6.5.

Theorem 8. *The tiling problem is undecidable even when restricted to tile sets that are deterministic by any two edges.*

Proof. A 2 × 2 tile set (see the definition of an $n \times n$ tile set in Section 2.1) of a 4-way deterministic tile set is not only 4-way deterministic but also deterministic by opposite edges. Therefore, the claim follows from the previous theorem. □

6.2. Layer 1: The aperiodic tile set

The aperiodic tile set of Kari and Papasoglu is used as the first layer of the new sandwich tile set. This 4-way deterministic tile set can be mapped homomorphically onto Robinson's tile set and therefore it is possible to use the patterns of red squares and blue squares of Robinson's original tile set in the rest of the proof. The aperiodic tile set is used to admit only non-periodic valid tilings in which the seed tile is contained infinitely many times.

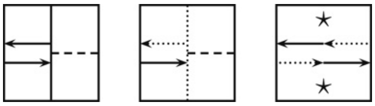
6.3. Layer 2: Identifying the free areas

A tile within a red square is said to be located on a *free column* of the red square, if there are no smaller red squares above or below it within the red square. Likewise, a tile within a red square is said to be located on a *free row* of the red square, if there are no smaller red squares within the red square to the right or to the left from its position. A tile within a red square is said to be *free*, if it is located on both a free row and a free column. For example, the free tiles of a $(2^6 + 1) \times (2^6 + 1)$ red square are shown in Fig. 22. The aperiodic tile set will be modified so that it is possible to determine whether a given tile within a red square is located on a free area or not.

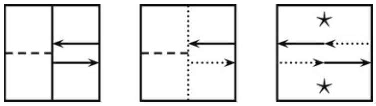
A row is said to be *red*, if it consists of tiles having (only) red horizontal arrows. Similarly, a row is said to be *blue*, if it consists of tiles having (only) blue horizontal arrows. Likewise, a column of tiles is said to be *red* or *blue* if it consists of tiles having red or blue vertical arrows, respectively. It needs to be noted that a row within a red square can be free only if it is a part of a blue row.

To identify the free rows (construction is only rotated to identify free columns), the aperiodic tile set is now modified as follows:

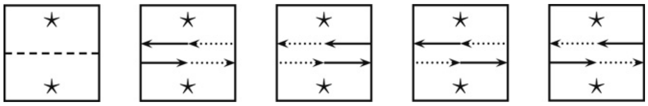
(1) The tiles on the west edges of red squares are paired with the tiles in Fig. 23(a).
(2) The tiles on the east edges of red squares are paired with the tiles in Fig. 23(b).



(a) The tiles to be paired with west edge red double arrows.



(b) The tiles to be paired with east edge red double arrows.



(c) The tiles to be paired with any vertical red single arrows.

Fig. 23. The tiles to be paired with the tiles on blue rows and red columns. Expression $\star$ denotes either solid or dotted vertical line and it is the same for both the north edge and the south edge.

(3) The tiles on red vertical single arrows are paired with the tiles in Fig. 23(c).
(4) The rest of the tiles are paired with such tiles that the colors originating from tiles in Fig. 23(a), (b) and (c) can intersect unchanged.

The idea of the tiles in Fig. 23 is that in a valid tiling a tile within a red square is located on a free row if, and only if, it contains a horizontal dashed line. If the tile is located on a blue row but not on a free area, then it must have a pair of horizontal arrows traveling in opposite directions. Each of these arrows is either solid or dotted. On a red column a solid arrow is changed to a dotted arrow and vice versa.

Every red column contains a single vertical line which is either solid or dotted. These vertical lines are used to determine how a dashed line representing a free row is swapped to the two arrows on the edge of a red square. Practically speaking, the vertical lines are used to identify nondeterministically, how many red columns there are between two nearest red squares of equal size. If there is an even number of red columns between the two squares, then both the east edge of the leftmost square and the west edge of the rightmost square should contain a solid vertical line. If there is an odd number of red columns between the two squares, then both the east edge of the leftmost square and the west edge of the rightmost square should contain a dotted vertical line.

On the edge of a red square a dashed line representing a free row is replaced with a leftward arrow and a rightward arrow. The replacement is determined by the vertical line (which is either solid or dotted) on that red column. If an east edge contains a solid vertical line, the rightward arrow (which is emitted away from the square) must be solid also. If an east edge contains a dotted vertical line, the rightward arrow must be dotted also. If a west edge contains a solid vertical line, the leftward arrow must also be solid. If a west edge contains a dotted vertical line, the leftward arrow must also be dotted. When the vertical lines are assumed to be of the same type on opposite edges of neighboring red squares, the arrow which is absorbed on a red edge is uniquely determined by the emitted arrow on the edge of the opposite red square.

The solid or dotted black arrows emitted at the ends of free rows of two neighboring red squares of equal size may intersect a vertical edge of another (larger) red square only exactly in the middle between the two squares. Let the two squares be of size $2^{2n}+1$ and let them be aligned so that both their horizontal edges are located on the same rows, the leftmost square has its east edge located on column $x=-2^{2n-1}$, the rightmost square has its west edge located on column $x=2^{2n-1}$ and the squares have free rows on a blue row in location $y=0$. Then on intervals $(-2^{2n-1},0)$ and $(0,2^{2n-1})$ there is an even number of red columns and the column $x=0$ is either red or blue. Furthermore, on row $y=0$ there are no vertical red square edges on intervals $(-2^{2n-1},0)$ and $(0,2^{2n-1})$. The only possibility on row $y=0$ to have a red vertical edge (i.e. a vertical red double arrow on Robinson's tiles) on interval $(-2^{2n-1},2^{2n-1})$ is in location $(x,y)=(0,0)$. Because both the subintervals contain an even number of red columns, the horizontal black arrows are tiled correctly in location $(0,0)$ using the tiles in Fig. 23(c) if there is no red vertical edge present. If there is a red vertical edge present in location $(0,0)$, then the horizontal black arrows are still drawn correctly using the third tile in Fig. 23(a) and (b).

It should be noted that there exists such a tiling in which there is a blue row which does not intersect any red squares. It follows that this row can therefore contain either a pair of horizontal arrows or a dashed line without causing a tiling error.

Lemma 9. *If a tile within a red square is located on a non-free row in a valid tiling, then it cannot contain a horizontal dashed line.*

Proof. If a tile on a non-free row contains a dashed line, the line would collide with an edge of a smaller red square. But this is not possible, because a dashed line coming from the outside of a square cannot collide with an edge of the square by the tiles in Fig. 23(a) and (b). □

Lemma 10. *If a tile within a red square is located on a free row in a valid tiling, then it must contain a dashed line.*

Proof. The arrows cannot be of the same type (solid or dotted) because then at the end of the free row the arrows would still be of the same type. A leftward arrow and a rightward arrow of the same type is not allowed inside the red square by the tiles in Fig. 23(a) and (b).

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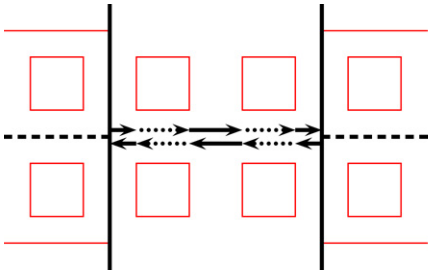
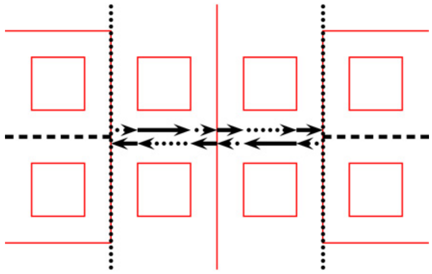


(a) Both the horizontal arrows are solid at the end of a free row if the area between the two squares contains an even number of red columns.



(b) The horizontal arrows are of different type at the end of a free row if the area between the two squares contains an odd number of red columns.

Fig. 24. Different tiles are used at the end of free rows depending on the (even or odd) number of red columns between the two red squares.

If the arrows are not of the same type, then they will still meet the red square edges on the east side and the west side as the same arrow type combination because there is an even number of red columns within the square and at every red column the arrow types are swapped. But the same arrow combinations arriving at the both edges is not possible because on the west side the leftward arrow and the rightward arrow must be dotted and solid, respectively, whereas the east side the leftward arrow and the rightward arrow must be solid and dotted, respectively, by the third tile in Fig. 23(a) and (b). □

By Lemmas 9 and 10, it is obvious that in a valid tiling a tile is located on a free row if, and only if, it has a horizontal dashed line. It remains to be shown that a valid tiling exists.

Lemma 11. *The modified tile set admits a valid tiling.*

Proof. A row which does not intersect with any red square edges can be tiled correctly. Therefore, it is enough to consider only rows that contain free tiles for some red squares. Consider row $y = 0$, which is assumed to contain free rows of red squares of size $2^{2n} + 1$.

Clearly, any free row itself within a square of size $2^{2n}+1$ can be tiled correctly by the tiles containing a horizontal dashed line. It is therefore sufficient to show that no tiling error is introduced on row $y = 0$ on the area between two neighboring red squares of size $2^{2n} + 1$ assuming that at the ends of a free row the dashed line is replaced with a suitable pair of arrows. Let the two neighboring squares be located so that the leftmost square has its east edge on column $x = -2^{2n-1}$ and the rightmost square has its west edge on column $x = 2^{2n-1}$.

If the area in between (i.e. columns on interval $(-2^{2n-1}, 2^{2n-1})$) contains an even number of red columns (as shown in Fig. 24(a)), the column $x = 0$ is blue and the row $y = 0$ can be tiled correctly by choosing the vertical lines to be solid on the edges of the red squares. The selection which is made between solid and dotted vertical lines on the east and west edges of red squares is "row invariant" because the selection will depend only on the number of red columns between the red squares.

If the area in between the two squares contains an odd number of red columns (as shown in Fig. 24(b)),the vertical lines are chosen to be dotted on the edges of the red squares. Then the east edge of the leftmost red square has a dotted rightward arrow and a solid leftward arrow. If the interval $(-2^{2n-1}, 2^{2n-1})$ does not contain a vertical red edge, the arrows are swapped an odd number of times on the interval using the tiles in Fig. 23(c) and the west edge of the rightmost square will have a solid rightward arrow and a dotted leftward arrow. These arrow combinations are allowed by the tiles in Fig. 23(a) and (b). Therefore, no tiling error is introduced if the interval $(-2^{2n-1}, 2^{2n-1})$ does not contain a vertical red edge and the vertical line components are chosen correctly on the edges of the red squares.

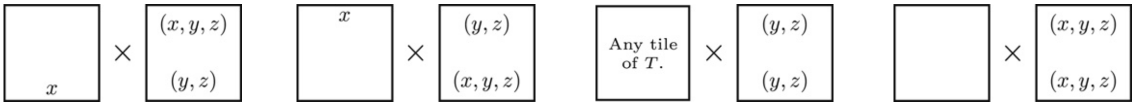
The only location on row $y = 0$ where the horizontal (solid or dotted) black arrows might meet a vertical red edge on interval $(-2^{2n-1}, 2^{2n-1})$ is on column $x = 0$. Because the arrows are swapped an even number of times on both intervals $(-2^{2n-1}, 0)$ and $(0, 2^{2n-1})$, the tile required on column $x = 0$ is exactly the third tile in Fig. 23(a) and (b) or the fourth tile in Fig. 23(c) (which are all the same). Therefore, no tiling error is introduced even if the interval $(-2^{2n-1}, 2^{2n-1})$ contains the only possible occurrence of a vertical red edge in the middle of the interval. □

The following theorem follows as a combination of the previous lemmas and their rotated counterparts:

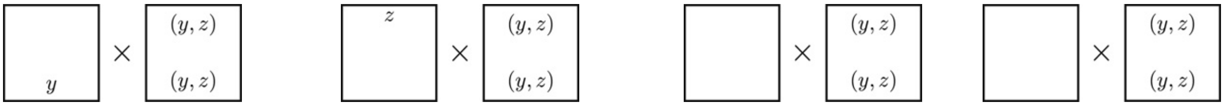
Theorem 12. *The 4-way deterministic aperiodic tile set of Kari and Papasoglu can be modified so that the set of tiles on free areas and the set of tiles on non-free areas are disjoint sets.*

6.4. Layer 3: Simulating the original tile set

Assume that the given instance for the tiling problem with a seed tile is the tile set T and the seed tile t. A tiling by the original tile set T is simulated within all the red squares. However, since larger red squares contain smaller red squares, the simulation area cannot be the entire square itself. Instead, the simulation corresponding the particular red square is done on the free areas.



(a) The tiles at the north border of a free area. (b) The tiles at the south border of a free area. (c) The tiles on free areas. (d) The tiles on non-free areas.



(e) The tiles on a red north edge next to a free area. (f) The tiles on a red south edge next to a free area. (g) The tiles on a red north edge next to a non-free area. (h) The tiles on a red south edge next to a non-free area.

Fig. 25. The tile construction on layers 3 and 4 to transfer vertical colors between the free areas inside a red square and between different red squares. Symbols x, y and z denote arbitrary colors of the given tile set T. Set T is assumed not to have blank color on any tile.

Lemma 13 (*Robinson [2]*). *For every $(4^n + 1) \times (4^n + 1)$ red square, the number of free columns is $2^n + 1$ and the number of free rows is $2^n + 1$.*

Lemma 13 states that the free area within a red square increases with respect to the size of the square. Hence, the tiling by the original tiles (on layer 3) can be arbitrarily large even when restricted to the free rows and free columns. Hence, it is enough to restrict the simulation only to free rows and free columns.

One copy of the seed tile can be forced to be located in the center of every red square (and therefore in the center of every simulation area) by pairing the seed tile with all blue cross tiles that are not part of a 3×3 blue square. Whether a cross tile is part of a 3×3 blue square is determined simply by using a 3×3 tile set version of the aperiodic tile set. The tiles that are used on this layer are the tiles of the original tile set T (which is assumed not to have blank as a color), the blank tile and some other tiles that have some sides colored blank and some colored with the colors of T. In Section 6.5 these tiles will be described in more detail and it will be shown to be possible to consider the area consisting of the free areas as a single continuous square.

6.5. Layer 4: Joining the free areas

To treat all the free areas within a red square as a single continuous area and to allow any colors at the edges of red squares while maintaining the 4-way determinism, a set of tiles is added to the original tile set and these tiles are paired with another set of tiles as shown in Fig. 25. Notice that the tiles in Fig. 25 forward only colors of the north and south edges. A similar construction is used to forward colors in the horizontal direction and the actual tile set is the combination of tiles in Fig. 25 and their counterparts for the horizontal direction.

Let f be a valid tiling of the aperiodic tile set. A tile $f(x, y)$ is said to be located on the *north border*, *east border*, *west border* or *south border* of a free area if a tile $f(x, y-1)$, $f(x-1, y)$, $f(x+1, y)$ or $f(x, y+1)$, respectively, belongs to the free area and the tile $f(x, y)$ does not. Collectively, the tiles that do not belong to a free area but are located horizontally or vertically next to the free area are said to be located on a *border* of the free area. The locations enumerated in Fig. 25 can be identified as disjoint subsets of the aperiodic tile set after the modifications of layer 2 (i.e. Section 6.3) and forming the 3×3 tile set of the new modified aperiodic tile set. By looking at the tiles in Fig. 25, it is obvious that the final tile set is 4-way deterministic.

What needs to be shown is that tiles force arbitrarily large squares to be tiled with the tiles of the original tile set. This is done by erasing colors at the free area borders and forwarding them to the next free area unless the color is adjacent to a red edge. At the red edges the color is erased and forwarded to the next red square. Another way of looking at it is that the final colors next to red edges are chosen nondeterministically for all red squares of the same size.

For example, at the north border of a free area the last color (i.e. x in Fig. 25) on layer 3 is erased and raised onto layer 4 to be transferred northwards. At the south border of another free area color x is lowered from layer 4 back to layer 3. The last colors next to red edges are chosen non-deterministically using the tiles in Fig. 25(e) and (f). The principle of forwarding colors is represented in Fig. 26.

It can be concluded that any color is allowed on layer 3 next to a red edge and the free areas can be considered as a single continuous area. Therefore the new tile set can simulate tiling of arbitrarily large squares by the original tile set and Theorem 7 follows.

7. Conclusions

It was shown that the tiling problem with Wang tiles remains undecidable even if the instances are 4-way deterministic tile sets (Theorem 7) or deterministic by any two edges (Theorem 8). It has been known that 4-way deterministic aperiodic tile sets exist [4], but undecidability of the tiling problem has been an open problem. The proof followed the same general structure as that of Robinson [2] but with many tedious technical details.

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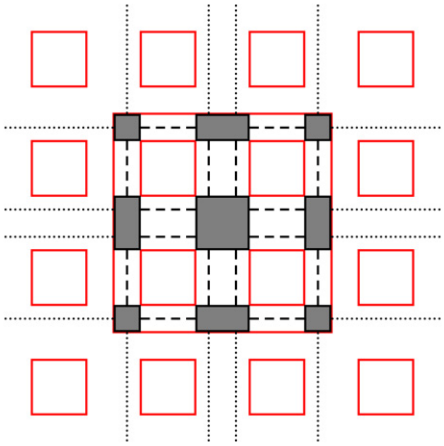

Fig. 26. Forwarding the colors from one free area to another and from one red square to another using the construction in Fig. 25. Dotted lines represent two-color vectors and dashed lines represent three-color vectors. Darkened rectangles represent free areas.

It is an open problem what kind of patterns can be "drawn" with 4-way deterministic tile sets. Is it possible to give some sort of characterization of the patterns that can be tiled with a 4-way deterministic tile set? It is quite straightforward to show that any finite pattern can be drawn with a 4-way deterministic tile set. Answering these two questions might give theoretical results on how to self-assemble, say, molecular scale electronic circuits, efficiently with very few errors. How complex do the tile sets have to be for each (infinite) pattern? Could there be a significantly simpler tile set for drawing a single diagonal line 4-way deterministically? The tiling which contains the diagonal line must be non-periodic but the tile set does not have to be aperiodic. How can the definition of (4-way) determinism be extended to the case of mathematical self-assembly in the most useful way?

Acknowledgments

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References

[1] R. Berger, The undecidability of the domino problem, Memoirs of the American Mathematical Society 66 (1966) 1–72.
[2] R.M. Robinson, Undecidability and nonperiodicity for tilings of the plane, Inventiones Mathematicae 12 (1971) 177–209.
[3] J. Kari, The nilpotency problem of one-dimensional cellular automata, SIAM Journal on Computing 21 (1992) 571–586.
[4] J. Kari, P. Papasoglu, Deterministic aperiodic tile sets, Geometric and Functional Analysis 9 (1999) 353–369.
[5] J. Kari, Reversibility and surjectivity problems of cellular automata, Journal of Computer System Sciences 48 (1994) 149–182.
[6] J. Kari, Theory of cellular automata: A survey, Theoretical Computer Science 334 (2005) 3–33.
[7] J. Kari, V. Lukkarila, Some undecidable dynamical properties for one-dimensional reversible cellular automata (in press).
[8] E. Winfree, Algorithmic self-assembly of DNA, Ph.D. Thesis, California Institute of Technology, 1998.
[9] L. Adleman, Q. Cheng, A. Goel, M.-D. Huang, D. Kempe, P.M. de Espanés, P.W.K. Rothemund, Combinatorial optimization problems in self-assembly, in: Proceedings 34th Annual ACM Symposium on Theory of Computing, 2002, pp. 23–32.
[10] G. Aggarwal, Q. Cheng, M.H. Goldwasser, M.-Y. Kao, P.M. de Espanés, R.T. Schweller, Complexities for generalized models of self-assembly, SIAM Journal on Computing 34 (2005) 1493–1515.
[11] E. Winfree, Self-healing tile sets, in: J. Chen, N. Jonoska, G. Rozenberg (Eds.), Nanotechnology: Science and Computation, Springer-Verlag, 2006, pp. 55–78.
[12] D. Soloveichik, M. Cook, E. Winfree, Combining self-healing and proofreading in self-assembly, Natural Computing 7 (2008) 203–218.
[13] J.E. Hopcroft, J.D. Ullman, Introduction to Automata Theory, Languages and Computation, Addison-Wesley, 1979.
[14] Z. Manna, Mathematical Theory of Computation, McGraw-Hill, 1974.
[15] H. Wang, Proving theorems by pattern recognition II, Bell Systems Technical Journal 40 (1961) 1–41.